

Task 2:

Phishing Email Detection & Awareness System

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Summary:

1. Executive summary
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3. Methodology
4. Email Analysis
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Executive Summary:

Email remains one of the most common attack vectors used by cybercriminals to compromise organizations and individuals. Phishing attacks exploit social engineering techniques to deceive users into revealing sensitive information, clicking malicious links, or downloading harmful attachments. As these attacks continue to evolve, organizations must combine technical security measures with user awareness to reduce the risk of successful compromise.

This report presents the results of a phishing email assessment conducted on publicly available email samples. The assessment focused on identifying common phishing indicators, including spoofed sender addresses, suspicious domains, malicious hyperlinks, urgent language, and other social engineering techniques. Each email was analyzed, classified according to its risk level (Safe, Suspicious, or Phishing), and evaluated based on its potential impact on users and organizations.

The objective of this assessment is to improve phishing awareness by explaining how phishing attacks operate, highlighting common warning signs, and providing practical recommendations to help users recognize and respond appropriately to suspicious emails.

Executive Summary:

The report also offers preventive measures that organizations can adopt to strengthen their overall email security posture.

Scope:

This assessment focused on the analysis of publicly available phishing email samples in order to identify common phishing indicators and evaluate the associated security risks. The assessment was conducted in a controlled learning environment using publicly accessible tools and datasets, without interacting with live email systems or executing malicious content.

In Scope

The following elements were included in the assessment:

- Analysis of phishing and legitimate email samples
- Inspection of sender email addresses and domains
- Examination of email headers
- Analysis of embedded URLs and hyperlinks
- Identification of social engineering techniques
- Classification of emails as Safe, Suspicious, or Phishing
- Assessment of potential business impact and security risks
- Development of user awareness and prevention recommendations

Out of Scope

The following activities were intentionally excluded from this assessment:

- Malware execution or attachment analysis
- Interaction with live phishing campaigns
- Active exploitation or penetration testing
- Email server configuration assessment
- Incident response or forensic investigation
- Analysis of private or confidential email communications

Methodology:

The phishing email assessment was conducted using a structured and non-intrusive approach to evaluate publicly available email samples. The objective was to identify common phishing characteristics, assess the associated risks, and develop practical recommendations for improving email security awareness.

Tool Used:

- Public phishing email samples
- Email Header Analyzer
- Terminal (Command Line Interface)
- Web Browser

The assessment followed a structured and non-intrusive methodology to evaluate phishing emails from both technical and user-awareness perspectives. The process began with the selection of publicly available phishing email samples representing common phishing scenarios. These samples provided a realistic basis for identifying the techniques commonly used by attackers to deceive recipients and compromise sensitive information.

Each email was examined by reviewing the sender's address, domain name, and email headers to identify signs of spoofing, domain impersonation, or other inconsistencies that could indicate a fraudulent source. Email headers were further analyzed using an Email Header Analyzer and command-line tools to verify message routing information and assess the authenticity of the sender.

The email content was then analyzed to identify common phishing indicators, including suspicious hyperlinks, requests for sensitive information, urgent or threatening language, unexpected attachments, and other social engineering techniques. Embedded URLs were inspected without accessing potentially malicious content, ensuring that the assessment remained within a safe, passive, and ethical scope.

Email Analysis:

To perform this assessment, three publicly available phishing email samples were selected from the Phishing Pot repository developed by rf-peixoto. These samples represent common phishing scenarios and provide a safe environment for learning and security analysis.

Each email was analyzed using a combination of technical tools, including an Email Header Analyzer, command-line utilities, and a web browser. The analysis focused on examining sender information, email headers, embedded links, and social engineering techniques to identify phishing indicators and assess the associated security risks.

The following sections present the analysis results for each email sample, supported by the corresponding commands, screenshots, and observations collected during the assessment.



Example 1: Command Line

```
pro@pro-HP-EliteBook-x360-1030-G4:~/Downloads/email$ rg "^(From:|To:|Subject:|Date:|Return-Path:|Reply-To:|Message-ID:)" sample-1017.eml
26:From: Magisches Schlankheitssystem ✂️✅ <otto-newsletter@newsletter.otto.de>
27:Subject: Individuelle Prognose für schnellen Gewichtsverlust ✅👥
28:Reply-To: reply_to@winner-win.art
29:To: phishing@pot
33:Date: Thu, 03 Aug 2023 04:14:33 +0200
36:Return-Path: return@winner-win.art
pro@pro-HP-EliteBook-x360-1030-G4:~/Downloads/email$ rg "^Received:" sample-1017.eml
1:Received: from CYXPR19MB8312.namprd19.prod.outlook.com (::1) by
4:Received: from DB8PR06CA0029.eurprd06.prod.outlook.com (2603:10a6:10:100::42)
9:Received: from DB8EUR05FT062.eop-eur05.prod.protection.outlook.com
19:Received: from qktfxthukmwfnksijbkrjxkzhstbswa.whstr8 (80.96.157.90) by
pro@pro-HP-EliteBook-x360-1030-G4:~/Downloads/email$ rg "SPF|DKIM|DMARC|Authentication-Results"
sample-1004.eml
14:Authentication-Results: spf=softfail (sender IP is 159.27.24.86)
18:Received-SPF: SoftFail (protection.outlook.com: domain of transitioning
sample-1025.eml
14:Authentication-Results: spf=none (sender IP is 89.144.9.91)
```

```
sample-1017.eml
7:Received-SPF: SoftFail (protection.outlook.com: domain of transitioning
```



```
pro@pro-HP-EliteBook-x360-1030-G4:~/Downloads/email$ rg "https?:://"
sample-1017.eml
154:<meta http-equiv="Content-Type" content="text/html; charset=utf-8"><title>Ket
toXplode</title> </head> <body centermargin="0" topmargin="0" marginwidth="0" m
arginheight="0" bgcolor="#eeeeee"> <center> <a href="https://t.co/eYVtqVunRC" ta
rget="_blank" style="color:#202020;text-decoration:none">age ist der Vorrat an K
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700px; font-weight:100; padding-top:5px; padding-bottom:5px; line-height:17px">
Aufgrund der extrem hohen Nachfr Beeilen Sie sich! </td> </tr> </table> <!--Preh
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164:<tr> <td align="center" style="background-color:#f8f3f0; padding-top:15px; f
ont-family:Helvetica, Arial, sans-serif; font-size:32px; line-height:42px; color
:#2a2a2a; font-weight:300; text-align:center;"> Individuelle Prognose für Ihr sc
hnelles Gewichtsmanagement </tr> </table> </td> </tr> <tr> <td align="center" st
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t="_blank" style="color:#f8f3f0; text-decoration:none;"> <table width="100%" border="0">
Discussion informelle et projets
```

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Example 1: Email Header:

MessageId	qAlAcXa.60981.504+=phishing@pot@winner-win.art
Created at:	8/3/2023, 3:14:33 AM GMT+1 (Delivered after 34 sec)
From:	Magisches Schlankheitssystem  <otto-newsletter@newsletter.otto.de>
To:	phishing@pot
Subject:	Individuelle Prognose für schnellen Gewichtsverlust  
SPF:	softfail with IP Unknown! Learn more
DKIM:	none Learn more
DMARC:	fail Learn more

#	Delay	From *	To *	Protocol	Time received
0	34 sec	DB8EUR05FT062.eop-eur05.prod.protection.outlook.com	→ DB8PR06CA0029.outlook.office365.com		8/3/2023, 3:15:07 AM GMT+1

Example 2: Command Line

```
pro@pro-HP-EliteBook-x360-1030-G4:~/Downloads/email$ rg "^(From:|To:|Subject:|Date:|Return-Path:|Reply-To:|Message-ID:)" sample-1025.eml
25:From: Microsoft account team ,_<no-reply@access-accsecurity.com>
26:Subject: Microsoft account unusual signin activity
27:To: phishing@pot
30:Date: Wed, 2 Aug 2023 03:34:32 +0000
31:Reply-To: solutionteamrecognizd03@gmail.com
35:Message-ID:
37:Return-Path: bounce@iustozncau.co.uk
pro@pro-HP-EliteBook-x360-1030-G4:~/Downloads/email$ rg "^Received:" sample-1025.eml
1:Received: from SA0PR19MB4444.namprd19.prod.outlook.com (::1) by
4:Received: from SN7P222CA0014.NAMP222.PROD.OUTLOOK.COM (2603:10b6:806:124::8)
9:Received: from SN1NAM02FT0020.eop-nam02.prod.protection.outlook.com
19:Received: from iustozncau.co.uk (89.144.9.91) by
pro@pro-HP-EliteBook-x360-1030-G4:~/Downloads/email$ rg "SPF|DKIM|DMARC|Authentication-Results"
sample-1004.eml
14:Authentication-Results: spf=softfail (sender IP is 159.27.24.86)
18:Received-SPF: SoftFail (protection.outlook.com: domain of transitioning
sample-1017.eml
14:Authentication-Results: spf=softfail (sender IP is 80.96.157.90)
17:Received-SPF: SoftFail (protection.outlook.com: domain of transitioning
sample-1025.eml
201:
pro@pro-HP-EliteBook-x360-1030-G4:~/Downloads/email$ rg "https?://" sample-1025.eml
201:
pro@pro-HP-EliteBook-x360-1030-G4:~/Downloads/email$
```

Example 2: Email header

MessageId	570d6a58-f5c2-476f-a67a-52d5ed0d1cc6@SN1NAM02FT0020.eop-nam02.prod.protection.outlook.com
Created at:	8/2/2023, 4:34:32 AM GMT+1 (Delivered after 1 sec)
From:	Microsoft account team _<no-reply@access-accsecurity.com>
To:	phishing@pot
Subject:	Microsoft account unusual signin activity
SPF:	none Learn more
DKIM:	none Learn more
DMARC:	permerror Learn more

#	Delay	From *	To *	Protocol	Time received
0	1 sec	SN1NAM02FT0020.eop-nam02.prod.protection.outlook.com	→ SN7P222CA0014.outlook.office365.com		8/2/2023, 4:34:33 AM GMT+1

Findings and Assessment:

The analysis of the two phishing email samples revealed several characteristics commonly associated with phishing campaigns. Although each email used different social engineering techniques, they shared the same objective: convincing recipients to trust the message and perform actions that could compromise sensitive information or organizational security.

One of the most significant findings was the use of spoofed or misleading sender identities. The analysis of the email headers and sender domains showed inconsistencies that suggested attempts to impersonate legitimate organizations. Such techniques increase the credibility of phishing emails and make them more likely to deceive recipients, particularly ¹¹ those who do not routinely verify sender information.

This finding was assessed as a High Risk due to its potential to facilitate credential theft and unauthorized access. The inspection of embedded hyperlinks also revealed suspicious URLs designed to redirect users to domains different from those displayed in the email content. This discrepancy is a common phishing technique used to lure victims toward fraudulent websites that imitate trusted services. If users fail to verify the destination before clicking, they may unknowingly disclose sensitive information such as usernames, passwords, or financial data. Therefore, this finding was also classified as ****High Risk**** because of its direct impact on confidentiality and account security. In addition, the content analysis identified several social engineering techniques, including urgent requests, warnings about account suspension, and messages encouraging immediate action. These psychological manipulation tactics are intended to create a sense of pressure, reducing the likelihood that recipients will carefully inspect the email before responding. Although these techniques do not exploit technical vulnerabilities, they significantly increase the probability of human error and were therefore assessed as a **Medium Risk**.

Conclusion:

Effective protection against phishing requires a combination of technical security controls and user awareness.

Organizations should implement email authentication mechanisms such as SPF, DKIM, and DMARC, deploy advanced email filtering solutions, and provide regular security awareness training to employees. By combining these preventive measures with proactive monitoring, organizations can significantly reduce the risk of successful phishing attacks and strengthen their overall cybersecurity posture.